7 Liquid **A** has a density of 1.42 g/cm^3

7 cm^3 of liquid **A** is mixed with 125 cm^3 of liquid **B** to make liquid **C**.

Liquid **C** has a density of 1.05 g/cm^3

Find the density of liquid **B**.
Give your answer correct to 2 decimal places.

Mass A = 1.42 × 7 = 9.94g.

1 mark
Density × volume

Volume C = 7 + 125 = 132cm^3.

Mass C = 1.05 × 132 = 138.6g.

Mass B = 138.6 – 9.94 = 128.66g.

1 mark
Find B mass;
divide by B volume

Density B = 128.66 ÷ 125 = 1.02928g/cm^3.

Density B = 1.03g/cm^3 (2 decimal places).

1 mark
1.03g/cm^3

..... g/cm^3

(Total for Question 7 is 3 marks)